



Intuitionistic Fuzzy Sets in J-CO-QL⁺

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Abstract. Intuitionistic Fuzzy Sets extend the classical notion of Fuzzy Sets, so as to represent “hesitation”: indeed, an item has both a membership degree and a non-membership degree, whose sum could be less than 1; the difference denotes the “hesitation” about the fact that the item belongs or not to the fuzzy set. Similarly, Intuitionistic Fuzzy Relations involve two domains.

Supposing that Intuitionistic Fuzzy Sets and Relations are provided as JSON data sets, is there a stand-alone tool to process them? This paper studies if the constructs currently provided by *J-CO-QL⁺* (the query language of the *J-CO* Framework) for managing fuzzy sets can actually deal with Intuitionistic Fuzzy Sets and Relations. The results will suggest how to extend *J-CO-QL⁺* to deal with classical and Intuitionistic Fuzzy Sets in an integrated way.

Keywords: Intuitionistic Fuzzy Sets and relations · Soft Querying on JSON data sets · Managing Intuitionistic Fuzzy Sets in J-CO-QL⁺

1 Introduction

An Intuitionistic Fuzzy Set (IFS) is an extension of the classical notion of Fuzzy Set (FS) proposed by Zadeh [16], so as to represent “hesitation”: an item of an IFS has both a membership degree and a non-membership degree, whose sum could be less than 1; the difference denotes the “hesitation” about the fact that the item belongs or not to the fuzzy set. The classical notion of relation can be generalized as Intuitionistic Fuzzy Relation (IFR) as well.

Supposing that *JSON* data sets are provided as IFSs and IFRs, is there a stand-alone tool to process them? At University of Bergamo (Italy), we are developing the *J-CO* Framework: it is a pool of software tools whose goal is to allow analysts to integrate and query *JSON* data sets, possibly stored within *JSON* document stores, like *MongoDB* (the framework was inspired by our previous work [5]); The *J-CO-QL⁺* query language supports the evaluation of membership degrees of documents to fuzzy sets (this idea was inspired by our previous work [6]). In this paper, we study if and how *J-CO-QL⁺* is currently able to manage IFSs and IFRs. The outcomes of this study will suggest how to extend (as a future work) *J-CO-QL⁺* to deal with FSs and IFSs in an integrated way.

The goal of this paper is not to discuss if IFSs are better than FSs or possibility theory [9] (this discussion is outside the scope of the paper); in contrast, it evaluates how far away $J\text{-CO-QL}^+$ is from providing a full support to IFSs.

The paper is organized as follows. Section 2 briefly introduces FSs (Sect. 2.1), basic concepts on IFSs and IFRs (Sect. 2.2) and a simple case study (Sect. 2.3). Section 3 studies if and how it is possible to deal with IFSs and IFRs within $J\text{-CO-QL}^+$. Finally, Sect. 4 discusses the lessons we learned towards a further extension of $J\text{-CO-QL}^+$ that addresses multiple models of FSs at the same time; it also draws the conclusions.

2 Background

In this section, we briefly present the background of our work, i.e., basic notions on FSs (Sect. 2.1) and on IFSs and IFRs (Sect. 2.2), practically explained (in Sect. 2.3) through an example.

2.1 Classical Fuzzy Sets

First of all, we report basic notions about classical Fuzzy Sets [16].

Let us denote by U a non-empty universe, either finite or infinite.

Definition 1. A fuzzy set $A \subseteq U$ is a mapping $A : U \rightarrow [0, 1]$. The value $A(x)$ is referred to as the *membership degree* of the x item to the A fuzzy set.

Therefore, a FS A in U is characterized by a membership function $A(x)$ that associates each item $x \in U$ with a real number in the interval $[0, 1]$; in other words, given x , the value $A(x)$ is the degree with which x is member of A (alternatively, the notation $\mu_A(x)$ is used, in place of $A(x)$). Thus, $A(x) = 1$ means that x fully belongs to A ; $A(x) = 0$ means that x does not belong to A ; $0 < A(x) < 1$ means that x belongs to A , but only partially. Consequently, FSs express vague or imprecise concepts on real-world entities, as well as they express vague or imprecise relationships among real-world entities.

Definition 2. Consider a universe U and two fuzzy sets A and B in U . The *union* of A and B is $S = A \cup B$, where $S(x) = \max(A(x), B(x))$ (alternatively written as $\mu_S(x) = \max(\mu_A(x), \mu_B(x))$). The *intersection* of A and B is $I = A \cap B$, where $I(x) = \min(A(x), B(x))$ (alternatively written as $\mu_I(x) = \min(\mu_A(x), \mu_B(x))$).

In this paper, U is the universe set of *JSON* documents. Thus, given a document $d \in U$, the membership degree $A(d)$ denotes the extent with which d belongs to the A FS (d represents a real-world entity or a relation).

2.2 Intuitionistic Fuzzy Sets and Relations

We consider the extension of FSs called Intuitionistic Fuzzy Sets [1]. Classical FSs relies on the “Axiom for Excluded Middle” [2]: given an item $x \in U$ and a FS A , given $A(x)$ its degree of membership, the degree of “non membership” to A is necessarily $1 - A(x)$; so, introducing a function $\nu_A(x)$ that explicitly represents the non-membership degree, $\nu_A(x) = 1 - \mu_A(x)$. But this assumption does not cope with the idea that gray zones can characterize practical situations: is a given symptom s_1 shown by a patient a clear evidence that the patient has a specific disease d_3 ? A physician would “feel” this, based on past experience and knowledge, that is, the physician is confident that the s_1 symptom denotes the d_3 disease with strength $\mu_{d_3}(s_1)$, but the strength $\nu_{d_3}(s_1)$ of the feeling that s_1 does not denote d_3 is such $\nu_{d_3}(s_1) < 1 - \mu_{d_3}(s_1)$, meaning that the physician “hesitates” in making a decision. In other words, the physician relies on intuitions, which are characterized by a degree of hesitation between the two opposite cases.

Definition 3. Given a universe U , an Intuitionistic Fuzzy Set (IFS) A in U is defined as a function $A : U \rightarrow [0, 1]_\mu \times [0, 1]_\nu$, where $[0, 1]_\mu$ is the domain of membership degrees and $[0, 1]_\nu$ is the domain of non-membership degrees. Thus, given an item $x \in U$, $A(x) = \langle \mu, \nu \rangle$, where μ and ν are, respectively, the membership degree of x to A (written also as $\mu_A(x) \in [0, 1]$) and the non-membership degree of x to A (written also as $\nu_A(x) \in [0, 1]$). For all items $x \in U$ and an Intuitionistic Fuzzy Set A in U , the following constraint must be satisfied: $A(x).\mu + A(x).\nu \leq 1$ (alternatively written as $\mu_A(x) + \nu_A(x) \leq 1$). Finally, given an item $x \in U$ and an IFS A in U , $\pi_A(x) = 1 - (A(x).\mu + A(x).\nu)$ is called the *hesitation* of x .

Given an IFS A in U , if for all items $x \in U$ it is $A(x).\mu + A(x).\nu = 1$, then A is a classical FS. Thus, classical FSs are included in the domain of IFSs.

Classical set-oriented operations can be adapted for IFSs.

Definition 4. Consider a universe U and two IFSs A and B in U .

The *union* of A and B is $S = A \cup B$, where

$$S(x) = \langle \max(A(x).\mu, B(x).\mu), \min(A(x).\nu, B(x).\nu) \rangle$$

(alternatively, it is possible to write $\mu_S(x) = \max(\mu_A(x), \mu_B(x))$ and $\nu_S(x) = \min(\nu_A(x), \nu_B(x))$).

The *intersection* of A and B is $I = A \cap B$, where

$$I(x) = \langle \min(A(x).\mu, B(x).\mu), \max(A(x).\nu, B(x).\nu) \rangle$$

(alternatively, it is possible to write $\mu_S(x) = \min(\mu_A(x), \mu_B(x))$ and $\nu_S(x) = \max(\nu_A(x), \nu_B(x))$).

As the reader can see, IFSs are extensions of FSs.

Intuitionistic Fuzzy Relations. The notion of Intuitionistic Fuzzy Relation [3, 8] was used to model medical knowledge in a decision-making application (see more in Sect. 2.3).

Definition 5. Consider two sets X and Y . An *Intuitionistic Fuzzy Relation* (IFR) R from X to Y is an IFS in $U = X \times Y$ ($R : X \times Y \rightarrow [0, 1]_\mu \times [0, 1]_\nu$), thus $R(\langle x, y \rangle) = \langle \mu, \nu \rangle$ (alternatively, $\mu_R(\langle x, y \rangle)$ and $\nu_R(\langle x, y \rangle)$).

In the literature, the following notation is often used: $R(X \rightarrow Y)$.

What happens if an IFS A is defined on X ? It is reasonable to “compose” A and R , so as to obtain a derived IFS B on Y .

Definition 6. Consider an IFR $R(X \rightarrow Y)$ and an IFS A in X . The *max-min-max* composition is denoted as $B = A \bullet R$, where B is an IFS in Y . Given an item $y \in Y$,

$$\begin{aligned} B(y).\mu &= \max_{x \in X}(\min(A(x).\mu, R(\langle x, y \rangle).\mu)), \\ B(y).\nu &= \min_{x \in X}(\max(A(x).\nu, R(\langle x, y \rangle).\nu)). \end{aligned}$$

Similarly, it is possible to define the composition of two IFRs.

Definition 7. Consider two IFRs $R(X \rightarrow Y)$ and $Q(Y \rightarrow Z)$, where X , Y and Z are three domains. The *max-min-max* composition of R and Q is denoted as $T = R \bullet Q$, where T is an IFR in $X \times Z$. Given an item $\langle x, z \rangle \in X \times Z$,

$$\begin{aligned} T(\langle x, z \rangle).\mu &= \max_{y \in Y}(\min(R(\langle x, y \rangle).\mu, Q(\langle y, z \rangle).\mu)), \\ T(\langle x, z \rangle).\nu &= \min_{y \in Y}(\max(R(\langle x, y \rangle).\nu, Q(\langle y, z \rangle).\nu)). \end{aligned}$$

In the literature, the notation of the two compositions is $B = R \circ A$ and $T = Q \circ R$ (in our opinion, this notation is a little bit anti-intuitive, because the order of terms is reversed with respect to the way they are composed).

2.3 Example: Representing Medical Knowledge

From [8], we borrow the example for which it is possible to apply IFRs, i.e., representing medical knowledge. Consider three domains: P is the domain of patients, S is the domain of symptoms and D is the domain of diseases. Medical knowledge is described by an IFR $K(S \rightarrow D)$: membership degrees denote the confidence of the physician as far as a given symptom denotes a given disease, as well as the non-membership degree expresses the non-confidence; Fig. 1 shows a sample instance of K , on the right-hand side.

When a patient is visited, symptoms shown by the patient are recorded; since the way they are reported or observed is imprecise, they are represented by an IFR called $O(P \rightarrow S)$; Fig. 1 shows a sample instance of O , on the left-hand side. Composing the two relations O and K , we obtain $PD = O \bullet K$, where PD is an IFR $PD(P \rightarrow D)$ that associates a patient to his/her potential diseases; Fig. 2 shows the instance of the PD IFR obtained by composing the instances of O and K shown in Fig. 1.

Based on this new relation, for each patient $p \in P$ we can build the classical FS D_p in the set D of diseases, which maps a disease $d \in D$ (that is possibly affecting p) to the membership degree, by means of the following membership function: $D_p(d) = PD(\langle p, d \rangle).\mu - PD(\langle p, d \rangle).\nu \cdot \pi_{PD}(\langle p, d \rangle)$. In words, the membership degree is compensated by the product of the non-membership degree by

O		K	
$P \times S$	$\langle \mu, \nu \rangle$	$S \times D$	$\langle \mu, \nu \rangle$
$\langle \text{John, Temperature} \rangle$	$\rightarrow \langle 0.7, 0.2 \rangle$	$\langle \text{Temperature, Flu} \rangle$	$\rightarrow \langle 0.6, 0.1 \rangle$
$\langle \text{John, Headache} \rangle$	$\rightarrow \langle 0.7, 0.1 \rangle$	$\langle \text{Temperature, COVID-19} \rangle$	$\rightarrow \langle 0.7, 0.2 \rangle$
$\langle \text{John, Cough} \rangle$	$\rightarrow \langle 0.5, 0.3 \rangle$	$\langle \text{Temperature, Typhoid} \rangle$	$\rightarrow \langle 0.4, 0.5 \rangle$
$\langle \text{Jane, Temperature} \rangle$	$\rightarrow \langle 0.0, 0.4 \rangle$	$\langle \text{Headache, Flu} \rangle$	$\rightarrow \langle 0.4, 0.5 \rangle$
$\langle \text{Jane, Headache} \rangle$	$\rightarrow \langle 0.4, 0.5 \rangle$	$\langle \text{Headache, COVID-19} \rangle$	$\rightarrow \langle 0.5, 0.3 \rangle$
$\langle \text{Jane, Cough} \rangle$	$\rightarrow \langle 0.2, 0.6 \rangle$	$\langle \text{Headache, Typhoid} \rangle$	$\rightarrow \langle 0.5, 0.2 \rangle$
		$\langle \text{Cough, Flu} \rangle$	$\rightarrow \langle 0.2, 0.3 \rangle$
		$\langle \text{Cough, COVID-19} \rangle$	$\rightarrow \langle 0.4, 0.5 \rangle$
		$\langle \text{Cough, Typhoid} \rangle$	$\rightarrow \langle 0.1, 0.8 \rangle$

Fig. 1. IFRs O (symptoms shown by patients) and K (medical knowledge).

the hesitation degree; in [15], the *normalized Hamming distance* is used as membership function for D_p , which is probably better but this issue is outside the scope of this paper. The final disease chosen for each patient p is the one having the greatest membership degree in D_p . Figure 2 shows, on the right-hand side, the D_p FSs obtained for patients John and Jane, whose symptoms are reported in the instance of the O IFR shown in Fig. 1.

3 Intuitionistic Fuzzy Sets and $J\text{-CO-QL}^+$

We now address the core contribution of the paper. First of all, we clearly define the problem addressed in the study.

Definition 8 - Problem 1. Suppose that a collection `SymptomsDiseases` of *JSON* documents represents the IFR denoted as K in Sect. 2.3; also suppose that a collection `PatientsSymptoms` of *JSON* documents represents the IFR called O in Sect. 2.3. By means of $J\text{-CO-QL}^+$, process these two collections as IFRs, so as to obtain, for each patient, the D_p FS and associate each patient with the most likely disease, on the basis of shown symptoms.

Currently, $J\text{-CO-QL}^+$ is undergoing the revision of statements and the extension of them with constructs to support evaluation of FSs; specifically, for each document d , it is possible to evaluate its membership degrees to several FSs. Consequently, since $J\text{-CO-QL}^+$ currently does not support IFSSs, in order to solve Problem 1, it is necessary to think in terms of a couple of FSs for each $\langle \mu, \nu \rangle$ pair: the membership degree to a FS called `R.Membership` denotes $R(d).\mu$; the membership degree to a FS called `R.NonMembership` denotes $R(d).\nu$ (where R is the name of the IFR).

$PD = O \bullet K$		D_P	
$P \times D$	$\langle \mu, \nu \rangle$	John: D	μ
$\langle \text{John, Flu} \rangle \rightarrow$	$\langle 0.6, 0.1 \rangle$	Flu $\rightarrow$	0.56
$\langle \text{John, COVID-19} \rangle \rightarrow$	$\langle 0.7, 0.2 \rangle$	COVID-19 $\rightarrow$	0.68
$\langle \text{John, Typhoid} \rangle \rightarrow$	$\langle 0.5, 0.2 \rangle$	Typhoid $\rightarrow$	0.44
$\langle \text{Jane, Flu} \rangle \rightarrow$	$\langle 0.2, 0.4 \rangle$	Jane: D	μ
$\langle \text{Jane, COVID-19} \rangle \rightarrow$	$\langle 0.4, 0.4 \rangle$	Flu $\rightarrow$	0.04
$\langle \text{Jane, Typhoid} \rangle \rightarrow$	$\langle 0.4, 0.5 \rangle$	COVID-19 $\rightarrow$	0.32
		Typhoid $\rightarrow$	0.35

Fig. 2. IFR $PD = O \bullet K$ and classical fuzzy sets D_P .

3.1 $J\text{-CO-QL}^+$ Data Model and Execution Model

By means of $J\text{-CO-QL}^+$, it is possible to write complex queries or scripts, which are able to process collections of *JSON* documents taken from a *JSON* database like *MongoDB*. A query $q = (i_1, \dots, i_n)$ is a sequence of n (with $n > 0$) instructions i_j (with $1 \leq j \leq n$). Each instruction i_j receives a *query-process state* $s_{(j-1)}$ and produces an output query-process state s_j .

A query-process state is a tuple $s = \langle tc, IR, DBS, FO, JSF \rangle$. Specifically, tc is called *temporary collection* and contains the current collection of *JSON* documents to process; IR is a local and volatile database, where the query can temporarily save *Intermediate Results*; DBS is the set of *database descriptors*, so as to connect to external databases; FO is the pool of *fuzzy operators*, which are used to evaluate membership degrees of documents to FSs (see Sect. 3.2); JSF is the set of user-defined JavaScript functions, defined to empower the query with additional computational capabilities (see [12]).

A query works on collections of *JSON* documents, which can be either retrieved from an external source or database, or generated as temporary collection by previous instructions. So, the basic item to process is a *JSON* document d , as defined by the standard [7]. However, $J\text{-CO-QL}^+$ gives a special meaning to fields whose name begins with the $\sim$ character (which are still compliant with *JSON* naming rules for fields). Specifically, the $\sim\text{fuzzysets}$ field behaves as a key/value map: it is a nested document where the field name corresponds to a FS name, while the value of the field is the membership degree of d to the FS. Furthermore, the $\sim\text{geometry}$ field denotes the geometry of the real-world entity described by the document (in this paper, we do not make use of this field; the interested reader can refer to [4, 14]).

<pre>{ "Symptom" : "Temperature", "Disease" : "Flu", "Confidence" : 0.6, "NonConfidence": 0.1 }</pre>	<pre>{ "Patient" : "John", "Symptom" : "Temperature", "Confidence" : 0.7, "NonConfidence": 0.2 }</pre>
(a) Collection SymptomsDiseases.	(b) Collection PatientsSymptoms.

Fig. 3. Sample documents in the source collections.

3.2 *J-CO-QL*⁺ Script

Suppose that two collections are stored in a *MongoDB* database called *MyDB*. The first collection is called *SymptomssDiseases*: its documents represent medical knowledge, thus they correspond to items in the *K* IFR presented in Sect. 2.3 and shown in the right-hand side of Fig. 1. Figure 3a shows an example document in this collection: notice the fields named *Confidence* and *NonConfidence*, by means of which physicians express their confidence/non-confidence (which are going to become membership/non-membership to an IFR, but for physicians it is better to think in terms of confidence/non-confidence) for a symptom to denote a given disease.

The second collection is named *PatientsSymptoms* and its documents represent medical observations (the *O* IFR introduced in Sect. 2.3 and shown in the left-hand side of Fig. 1). Figure 3b shows an example document: the reader can see the presence of the fields named *Confidence* and *NonConfidence* too.

The script we wrote to address Problem 1 is reported in Listings 1 and 2; it is necessary to break it in two distinct parts, due to its length.

- The *USE DB* instruction on line 1 connects the query process to the *MongoDB* database called *MyDB*, managed by a server running on the local machine. It changes the *DBS* member in the query-process state; the other members (included the temporary collection) remain empty.

Listing 1 *J-CO-QL*⁺ script, Part 1.

```

1. USE DB MyDB ON SERVER MongoDB 'http://127.0.0.1:27017';

2. JOIN OF COLLECTIONS PatientSymptom@MyDB AS O, SymptomsDisease@MyDB AS K
CASE WHERE .O.Symptom = .K.Symptom;

3. CREATE FUZZY OPERATOR Fuzzify
PARAMETERS Membership TYPE FLOAT
PRECONDITION Membership >= 0 AND Membership <= 1
EVALUATE Membership
POLYLINE [ (0, 0), (1, 1) ];

4. FILTER
CASE WHERE WITH .K.Confidence, .K.NonConfidence, .O.Confidence, .O.NonConfidence
GENERATE
CHECK FOR
FUZZY SET K_Membership USING Fuzzify(.K.Confidence),
FUZZY SET K_NonMembership USING Fuzzify(.K.NonConfidence),
FUZZY SET O_Membership USING Fuzzify(.O.Confidence),
FUZZY SET O_NonMembership USING Fuzzify(.O.NonConfidence);

5. FILTER
CASE WHERE KNOWN FUZZY SETS K_Membership, K_NonMembership,
O_Membership, O_NonMembership
GENERATE
CHECK FOR
FUZZY SET Pair_Membership USING O_Membership AND K_Membership,
FUZZY SET Pair_NonMembership USING O_NonMembership OR K_NonMembership
BUILD { .Patient : .O.Patient,
.Symptom : .O.Symptom,
.Disease : .K.Disease }
KEEPING FUZZY SETS Pair_Membership, Pair_NonMembership;

6. GROUP
PARTITION WITH .Patient, .Disease
BY .Patient, .Disease INTO .Items;

```

- The JOIN instruction on line 2 has a twofold goal: it retrieves the two collections and joins their documents, aliased as O (the **PatientsSymptoms** collection) and as K (the **SymptomsDiseases** collection) for simplicity; as specified by the WHERE selection condition, for each pair of source documents o (from the O collection) and k (from the K collection), a new document is built if o and k has the same value for the **Symptom** fields. The output document contains two fields, named O and K, which contain the source paired documents o and k , respectively. Figure 4a shows a sample document, which is obtained by pairing the two documents reported in Figs. 3b and 3a.

This way, we can say that the structural part of the IFR composition $O \bullet K$ (see Sect. 2.3) has been done; nevertheless, we have not managed membership/non-membership yet.

- So as to deal with FSs, line 3 defines a “fuzzy operator” called **Fuzzify**, which will be used for evaluating membership degrees to FSs. It receives a single parameter called **Membership**; the PRECONDITION imposes that its value must be between 0 and 1 (otherwise, an exception is raised).

The EVALUATE expression says that the value of the **Membership** parameter is used as x -axis value for the linear function described by the POLYLINE clause (in [10,11,13] the reader can see that complex polylines can be specified as membership functions). The corresponding y -axis value is the membership degree returned by the operator.

Listing 2 *J-CO-QL+* script, Part 2.

```

7. CREATE FUZZY OPERATOR toClassical
  PARAMETERS
    Membership      TYPE FLOAT,
    NonMembership    TYPE FLOAT
  PRECONDITION Membership >= 0 AND Membership <= 1 AND
    NonMembership >= 0 AND NonMembership <= 1
  EVALUATE Membership - NonMembership*(1 - Membership - NonMembership)
  POLYLINE [ (0, 0), (1, 1) ];

8. FILTER
  CASE WHERE WITH ARRAY .Items
  GENERATE
    CHECK FOR
      FUZZY SET PD_Membership
        USING Fuzzify(MAX_ARRAY(.Items,NUMERIC,DOCUMENTS,~fuzzysets.Pair_Membership)),
      FUZZY SET PD_NonMembership
        USING Fuzzify(MIN_ARRAY(.Items,NUMERIC,DOCUMENTS,~fuzzysets.Pair_NonMembership)),
      FUZZY SET D_of_P
        USING toClassical(MEMBERSHIP_OF(PD_Membership), MEMBERSHIP_OF(PD_NonMembership))
    BUILD { .Patient : .Patient,
      .Disease : .Disease }
    KEEPING FUZZY SETS D_of_P;

9. GROUP
  PARTITION WITH .Patient
  BY .Patient INTO .Diseases DROP GROUPING FIELDS
  SORTED BY .~fuzzysets.D_of_P TYPE NUMERIC DESC;

10. EXPAND
  UNPACK WITH .Diseases
  ARRAY .Diseases TO .Disease;

11. FILTER
  CASE WHERE .Disease.position = 1
  GENERATE
    CHECK FOR FUZZY SET D_of_P USING Fuzzify(.Disease.item.~fuzzysets.D_of_P)
    BUILD { .Patient : .Patient,
      .Disease : .Disease.item.Disease,
      .Confidence : MEMBERSHIP_OF(D_o_P) }
    DEFUZZIFY;

12. SAVE AS Diseases_Of_Patients@MyDB;

```

- Lines 4 and 5 actually evaluate FSs on documents.

The **FILTER** instruction on line 4 evaluates the membership degrees to four FSs: **K.Membership** and **K.NonMembership** are obtained by means of the **Fuzzify** fuzzy operator in the **USING** clauses, moving from the **Confidence** and **NonConfidence** fields coming from the **K** input collection (thus, their membership degrees correspond to μ_K and ν_K); similarly, the **O.Membership** and **O.NonMembership** FSs represent μ_O and ν_O , resp... Line 5 actually computes the *min* (respectively *max*) value of the membership (respectively, non-membership) of each pair, by obtaining the membership degree to the **Pair.Membership** (resp. **Pair.NonMembership**) FS. The final **BUILD** block simplifies the document structure, keeping only the **Pair.Membership** and **Pair.NonMembership** FSs.

- Line 5 computes the inner part of equations in Definition 7; to complete their computation, it is necessary to aggregate all documents associating the same patient to the same disease, through different symptoms. The **GROUP** instruction on line 6 does that: documents having the same values for the **Patient** and **Disease** fields are grouped together; all grouped documents are put into an array field called **Items**.

To complete the process, it is necessary to continue with the second part of the script, which is reported in Listing 2.

- Line 7 creates a second fuzzy operator, called **toClassical**: its goal is to combine membership and non-membership degrees to obtain one single membership degree for a classical FS (to compute the D_p FSs introduced in

```
{
  "K" : {
    "Disease"      : "Flu",
    "Symptom"      : "Temperature",
    "Confidence"    : 0.6,
    "NonConfidence" : 0.1,
  },
  "O" : {
    "Patient"      : "John",
    "Symptom"      : "Temperature",
    "Confidence"    : 0.7,
    "NonConfidence" : 0.2
  }
}
```

(a) Sample document after line 2.

```
{
  "Patient"      : "John",
  "Disease"      : "Flu",
  "~fuzzysets" : {
    "D_of_P" : 0.56
  }
}
```

(b) Sample document after line 8.

```
{
  "Patient" : "John",
  "Disease" : {
    "position" : 1,
    "item" : {
      "Disease" : "Flu",
      "~fuzzysets" : {
        "D_of_P" : 0.56
      }
    }
  }
}
```

(c) Sample document after line 10.

```
{
  "Patiente" : "John",
  "Disease" : "Flu",
  "Confidence" : 0.56
}
```

(d) Sample document after line 8.

Fig. 4. Sample documents in the temporary collections.

Sect. 2.3). Notice that the operator receives two parameters, **Membership** and **NonMembership**; the **EVALUATE** clause combines them, and the resulting value is returned as membership degree by the operator (see again, the simple polyline defined in the **POLYLINE** clause).

- The **FILTER** instruction on line 8 has a twofold goal. First of all, it completes the computation of membership/non-membership degrees for the IFR composition; this is done by computing the membership degrees to the FSs named **PD_Membership** and **PD_NonMembership**, for which in the **USING** clauses the **Fuzzify** fuzzy operator is called with the value obtained by the **MAX_ARRAY** and **MIN_ARRAY** aggregation functions, which provide the maximum (respectively, minimum) membership degrees to the **Pair_Membership** (respectively, **Peir_NonMembership**) FS, by directly accessing the `~fuzzysets` field of the documents in the **Items** array; in fact, only top-level `~fuzzysets` fields are implicitly managed as membership degrees to FSs by *J-CO-QL+*, but they are regular fields, so can be explicitly accessed by the query. The resulting membership degrees to the **PD_Membership** and **PD_NonMembership** FSs represent μ_{PD} and ν_{PD} , respectively (see the left-hand side of Fig. 2).

The second goal is to obtain the membership degree to the D_p classical FS, called **D.of_P**: this is done by calling the **toClassical** fuzzy operator, providing, as actual parameters, the membership degrees to the **PD_Membership** and **PD_NonMembership** FSs.

The final **BUILD** block further simplifies the structure of documents, keeping only the **Patient** and **Disease** fields and the **D.of_P** FS (see the sample document in Fig. 4b).

- The last step to do is to select, for each patient, the disease with the highest membership degree. On line 9, documents are grouped on the basis of the **Patient** field, so as to have one single document for each patient; grouped documents are sorted in reverse order of membership degree to the **D.of_P** FS (notice that it is necessary again to explicitly refer to the `~fuzzysets` field inside the grouped documents).

The **EXPAND** instruction on line 10 unnests again all grouped documents: this way, as it is possible to see in Fig. 4c, the novel **Disease** field contains both the unnested item (the **item** field) and the **position** field, denoting the position occupied by the item in the original array.

This structure is suitable to select (line 11) only documents whose field **.Disease.position** is 1, meaning that the document occupied the first position in the source array. For those documents, the membership to the **D.of_P** FS is evaluated again by using the former membership degree (i.e., the maximum one) by explicitly accessing for the third time the nested `~fuzzysets` field; finally the **BUILD** block creates the final structure of output documents (see an example in Fig. 4d), which loses (see the **DEFUZZIFY** keyword) membership degrees, because the **Confidence** field is generated with the membership degree of the **D.of_P** FS.

The final line 12 saves the last temporary collection into the *MongoDB* database called **MyDB**, with name **Diseases_Of_Patients**.

4 Learned Lessons and Conclusions

Intuitionistic Fuzzy Sets and Relations seem to have a strong potential to represent intuitionistic knowledge and to help processing data with uncertainty when events are judged by humans. Although in the literature it is possible to find many papers that describe their exploitation in practical applications, their adoption is limited by the lack of general-purpose tools.

In Sect. 3, we studied if and how it is possible to adopt $J\text{-CO-QL}^+$ and its fuzzy constructs to compose two intuitionistic fuzzy relations and choose the most likely option for a given item in a domain. We demonstrated that it is possible to process IFRs, which is good news. Moreover, we saw that the process is tricky, because a specific support is missing. Synthetically, we have learned the following lessons, to pursue as future work.

- (i) Managing membership and non-membership degrees as two independent classical fuzzy sets is not the best solution, because they must be dealt with in an integrated way; thus, we guess that a different and polymorphic structure for the special `~fuzzysets` field is necessary, so as to manage classical and intuitionistic fuzzy sets in a unified way.
- (ii) Novel constructs are necessary within $J\text{-CO-QL}^+$, once the data model is adapted, to deal with classical and intuitionistic fuzzy sets at the same time. In particular, novel constructs to transform intuitionistic to classical (and vice versa) fuzzy sets are necessary.
- (ii) Fuzzy aggregators over arrays of nested documents with membership degrees to fuzzy sets are currently missing in $J\text{-CO-QL}^+$; we are aware of this lack; then, it becomes mandatory to define them in such a way they can deal both with classical and intuitionistic fuzzy sets.

The $J\text{-CO}$ Framework is available on a public GitHub repository¹.

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¹ Github repository: <https://github.com/JcoProjectTeam/JcoProjectPage>.

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